

Project Overview

Project

1. Problem Statement

The company wants to implement an internal AI chatbot system for multiple departments (HR, Exim, Finance, IT). The goal is to provide employees with private, document-grounded answers to queries, while maintaining data privacy and scalability.

2. Goals & Objectives

- Deploy a **multi-department chatbot system** using AnythingLLM concepts.
 - Ensure **privacy-first architecture** with self-hosted or custom LLMs.
 - Build a **frontend interface** that supports chat, document upload, and workspace switching.
 - Define the **application flow** between frontend, backend, and LLM models.
-

3. Key Features & Requirements

Backend Infrastructure

- **LLM hosting:** GPU/CPU servers or cloud.
- **Vector database:** For embeddings (LanceDB, Pinecone, Weaviate).
- **API layer:** REST/GraphQL endpoints for chat, upload, workspace management.
- **Authentication:** Role-based access (Admin, Manager, Employee).

Frontend Requirements

- **Frameworks:** React, Next.js, or Vue.
 - **Chat interface:** Conversational UI with history.
 - **Workspace selector:** Switch between HR, Exim, Finance.
 - **Document upload:** Drag-and-drop or file picker.
 - **Role-based login:** Authentication system.
 - **Analytics dashboard:** Track usage and performance.
-

4. Application Flow

1. Frontend Query

- Employee types query in chat UI.
- Request sent via API to backend.

2. Backend API

- Authenticates user.
- Identifies workspace context (HR, Exim, Finance).

3. RAG Pipeline

- Query converted to embeddings.
- Vector DB retrieves relevant documents.
- Docs + query passed to LLM.

4. LLM Role

- Generates natural language response using context.
- Ensures accuracy and conversational tone.

5. Response Delivery

- Backend sends response to frontend.
 - Frontend displays answer with references.
-

5. Frontend Development Steps

- **UI Design:** Wireframes for chat, workspace navigation, document upload.
 - **Authentication:** Implement login with JWT/OAuth.
 - **Chat Component:** Input box, streaming responses, conversation history.
 - **Workspace Management:** Sidebar/dropdown for switching.
 - **Document Upload:** File upload UI with progress feedback.
 - **API Integration:** Connect to `/chat`, `/upload`, `/workspaces`.
 - **Analytics Dashboard:** Query volume, FAQs, performance metrics.
-

Final Note

The company can confidently build a **modular, scalable chatbot system** using AnythingLLM concepts. The frontend developer's role is clear: design and implement the user interface, connect to backend APIs, and ensure usability across departments. Backend choices (Ollama, LangChain, etc.) are optional.